**Table 17.1** Classification of algae based on pigments

| Class | Example | Types of Pigments | Reserve food material |
|---|---|---|---|
| Bluegreen algae (Cyanophyceae) | *Ocillatoria* | Phycocyanin | Cyanophycean Starch |
| Green algae (Chlorophyceae) | *Chylamydomonas* | Chlorophyll | Starch |
| Brown algae (Phaeophyceae) | *Laminaria* | Fucoxanthin | Laminarian starch and Manitol |
| Red algae (Rhodophyceae) | *Polysiphonia* | Phycoerythirin | Floridian Starch |

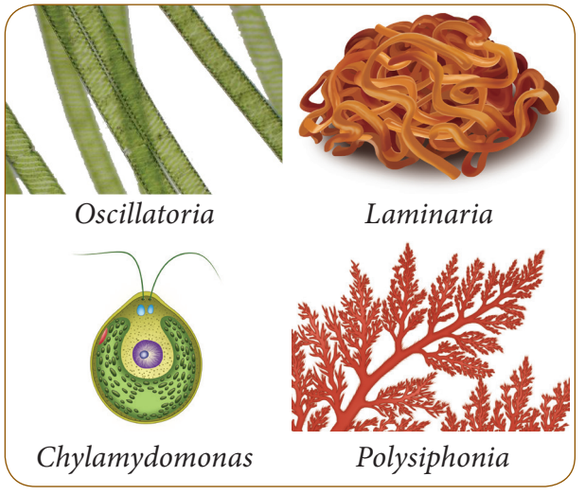



**Figure 17.1** Algae

## 17.1.2 Economic importance

**Food**

Algae are consumed as food by people in Japan, England and also in India. E.g. *Ulva, Spirulina, Chlorella* etc. Some algae are used as food for domestic animals. E.g. *Laminaria, Ascophyllum.*

**Agriculture**

Some of the blue green algae are essential for the fixing of atmospheric nitrogen into the soil, which increases the fertility of the soil. E.g. *Nostoc, Anabaena.*

**Agar Agar**

Agar agar is extracted from some red algae, namely *Gelidium* and *Gracillaria*. It is used to prepare growth medium in laboratories.

**Iodine**

Iodine is obtained from brown algae like *Laminaria*.

**Space travel**

*Chlorella pyrenoidosa* is used in space travel to get rid of $CO_2$ and to decompose human wastes.

**Single Cell Protein (SCP)**

Some of the single cell algae and blue green algae are used to produce protein. E.g. *Chlorella, Spirulina.*

## 17.2 Fungi

Fungi (Singular – Fungus) belongs to thallophyta. Its plant body is not differentiated into root, stem, and leaves. The plant body of fungus consists of filament like structures called **hyphae**. Several hyphae are arranged in the form of network called **mycelium**. The cells of fungi are multicellular and eukaryotic. Some species of fungi like yeast are unicellular and eukaryotic. Cell wall of fungi is made up of a chemical substance called chitin.

The reserve food materials of fungi are glycogen and oil. They have no starch because they have no chlorophyll pigments. So, they are heterotrophs. Heterotrophs are of three types namely, parasites, saprophytes and symbionts.

Some species of fungus live as parasites. They absorb food from the living organisms with the help of special root called haustoria. E.g. *Cercospora personata*. It affects groundnut plants and cause Tikka disease.

DO YOU KNOW?

The branch of study of fungus is called **mycology**.



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